Disambiguating Suicidal Intent: Mitigating Shortcut Learning and Catastrophic Forgetting via LIER

Korea University COSE461 Final Project

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Abstract

Ambiguous intent detection remains a major challenge in crisis NLP because identical risk-related keywords may express either genuine harmful intent or harmless figurative language depending on context. We investigate shortcut learning and catastrophic forgetting in risky versus non-risky intent classification using RoBERTa and NLI-based models, constructing a custom dataset of approximately 3,120 examples spanning 20 ambiguous keywords across negation, figurative, and temporal categories. Through extensive experiments, we show that near-perfect in-distribution (ID) accuracy is misleading, as models frequently fail under out-of-distribution (OOD) evaluation. SHAP analysis confirms that baseline models rely on isolated keywords rather than contextual understanding. To address this, we evaluate keyword masking and counterfactual augmentation, and propose **Linguistically-Informed Experience Replay (LIER)**, a lightweight continual learning framework that combines NLI initialization with a structured synthetic buffer organized by linguistic phenomenon category. Our best model achieves an OOD macro-F1 of 0.7531, outperforming the RoBERTa fine-tuned baseline (0.4531) by approximately 30 percentage points. Monte Carlo (MC) Dropout analysis further reveals that shortcut-sensitive models produce highly confident misclassifications on negated expressions, underscoring the importance of contextual robustness for reliable safety-critical intent classification.

1 Introduction

"I want to die." Three words that could save a life or be the punchline of a joke. The difference lies entirely in context. As AI systems are increasingly deployed in mental health screening and crisis detection Ji et al. (2022), their failure to understand ambiguous language carries real consequences. Recent incidents highlight these stakes: the estate of Suzanne Adams, killed by her son in a murder-suicide, alleged that ChatGPT validated his paranoid delusions rather than challenging them Ruwitch (2025), and a UK inquest revealed that teenager Luca Cella Walker bypassed its safety filters by framing his suicide inquiry as research, after which the chatbot provided specific methods Jolly and Brown (2026). Yet despite growing urgency, the detection of *ambiguous* suicidal intent, where identical surface forms carry opposing meanings, remains critically underexplored Yates et al. (2017);

Coppersmith et al. (2018); Resnik et al. (2021). Current approaches fail on expressions like *"I want to die of laughter"*, incorrectly flagging them as suicidal Chancellor and De Choudhury (2020); Zirikly et al. (2019). More critically, even sophisticated fine-tuned models exhibit this same surface-level failure silently. We show that RoBERTa, despite achieving 95%+ accuracy, learns dangerous shortcuts and catastrophically forgets its original linguistic abilities after fine-tuning, misclassifying *"I don't want to die"* as risky.

We make four contributions: (1) a novel dataset of approximately 3,120 contrast pairs spanning 20 ambiguous keywords in the suicide and mental health domain; (2) SHAP-based evidence of shortcut learning in fine-tuned RoBERTa; (3) a systematic study revealing that catastrophic forgetting persists across multiple architectural interventions; and (4) **Linguistically-Informed Experience Replay (LIER)**, a continual learning framework that preserves linguistic abilities while maintaining competitive accuracy. Our central finding: *dataset composition matters more than model architecture* for safety-critical NLP.

2 Related Work

Mental Health and Crisis NLP. Early work on mental health NLP focused on depression and self-harm detection from social media using lexical and sentiment-based features Yates et al. (2017); Coppersmith et al. (2018); Chancellor and De Choudhury (2020). Shared tasks such as CLPsych Zirikly et al. (2019) revealed that even state-of-the-art models struggle to distinguish degrees of suicide risk. Our work differs by explicitly targeting ambiguous intent, where identical keywords carry opposing risk levels, a problem largely overlooked in prior work.

Shortcut Learning in NLP. Gururangan et al. (2018) demonstrated that NLP models exploit spurious correlations rather than genuine linguistic understanding. McCoy et al. (2019) further showed that fine-tuned BERT relies on syntactic heuristics over semantic reasoning. We extend these findings to crisis NLP, using SHAP to show that RoBERTa attends to subject pronouns and isolated risk keywords without understanding the surrounding context, a particularly harmful shortcut in safety-critical applications where the same keyword can carry opposing intent.

Catastrophic Forgetting and Continual Learning. Catastrophic forgetting (McCloskey and Cohen, 1989) occurs when fine-tuning overwrites previously learned knowledge. Kirkpatrick et al. (2017) proposed Elastic Weight Consolidation (EWC) to mitigate this via weight regularization, penalizing changes to weights important for prior tasks. However, Rolnick et al. (2019) and de Masson d'Autume et al. (2019) demonstrated that experience replay consistently outperforms regularization-based approaches for NLP tasks, as replay actively reminds the model of important patterns rather than passively constraining weight updates. Motivated by these findings, we adopt experience replay as the foundation of our proposed LIER approach.

3 Approach

3.1 Task Formulation

We formulate ambiguous suicidal intent detection as binary classification. Given sentence x containing ambiguous keyword $k \in \mathcal{K}$, the model predicts $y \in \{0, 1\}$ (non-risky, risky). The core challenge is that identical keywords yield opposing labels depending on context: *"I want to die of laughter"* ($y = 0$) versus *"I want to die tonight"* ($y = 1$).

3.2 RoBERTa: From Pretrained to Fine-Tuned

We first evaluate **pretrained RoBERTa** (Liu et al., 2019) without fine-tuning as a pure baseline, then fine-tune on our dataset by passing the `[CLS]` token $\mathbf{h} \in \mathbb{R}^{768}$ through a linear classification head:

$$\hat{y} = softmax(\mathbf{W}\mathbf{h} + \mathbf{b}) \tag{1}$$

trained with cross-entropy loss using AdamW (Loshchilov and Hutter, 2019) with learning rate 2×10^{-5}, batch size 16, and 3 training epochs (Devlin et al., 2019; Sun et al., 2019).

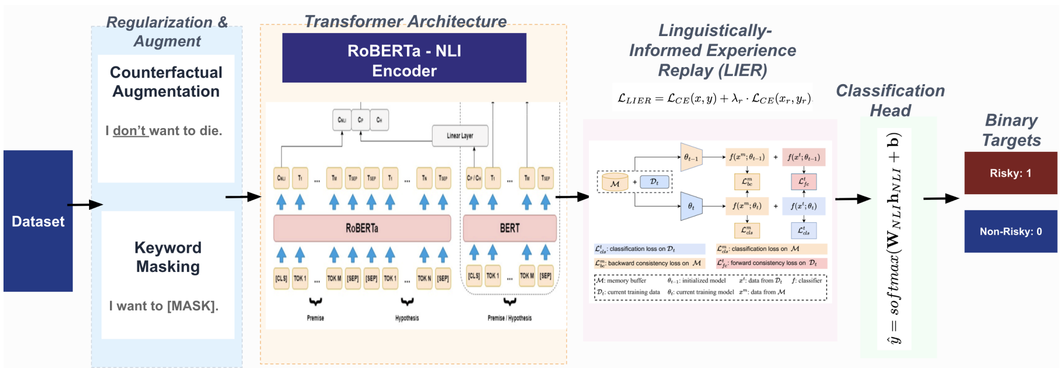


Figure 1: Overview of the proposed pipeline. The dataset is first processed through regularization and augmentation (keyword masking and counterfactual augmentation), then encoded via RoBERTa-NLI. LIER combines the main training loss with a linguistically-organized replay buffer $\mathcal{B}$, before passing through a binary classification head.

3.3 NLI-RoBERTa: Zero-Shot and Fine-Tuned

We evaluate `cross-encoder/nli-roberta-base` in two settings. NLI training encourages sentence-level semantic reasoning, directly relevant to distinguishing whether a risky keyword entails harmful intent or appears in a negated, figurative, or temporally shifted context (Dušek et al., 2023).

Zero-Shot. Given sentence x and candidate labels $\mathcal{Y} = \{risky, non-risky\}$, the NLI model computes entailment probability for each hypothesis h_y: *"This sentence expresses y intent"*:

$$p(y \mid x) = \frac{\exp(NLI(x, h_y))}{\sum_{y' \in \mathcal{Y}} \exp(NLI(x, h_{y'}))} \tag{2}$$

where $NLI(x, h_y)$ is the entailment score between premise x and hypothesis h_y. This requires no labeled data from our domain.

Fine-Tuned. We replace the NLI head with a binary classifier and fine-tune on our dataset:

$$\hat{y} = softmax(\mathbf{W}_{NLI}\mathbf{h}_{NLI} + \mathbf{b}) \tag{3}$$

where $\mathbf{h}_{NLI} \in \mathbb{R}^{768}$ is the CLS embedding from the NLI backbone. While zero-shot NLI handles negation and figurative language well, fine-tuning causes catastrophic forgetting (McCloskey and Cohen, 1989), motivating our proposed LIER framework.

3.4 Shortcut Learning Mitigation

A key challenge in our task is that high-risk keywords are not label-deterministic: the same word can indicate suicidal intent in one sentence but appear in a benign figurative, negated, or temporal context in another. Following prior work (Gururangan et al., 2018; McCoy et al., 2019), we evaluate two mitigation techniques.

Keyword Masking. Inspired by token-level noising methods (Wei and Zou, 2019; Feng et al., 2021), we randomly replace ambiguous risk keywords such as *die*, *kill*, and *jump* with `[MASK]` during training, discouraging the model from learning direct keyword-to-label shortcuts and encouraging reliance on surrounding context.

Counterfactual Augmentation. Motivated by Kaushik et al. (2020) and Gardner et al. (2020), we augment training data with the same risk keyword in both risky and non-risky contexts, such as *"I want to die tonight"* ($y = 1$) and *"I want to die of laughter"* ($y = 0$). This removes keyword reliability as a standalone label cue, forcing the model to learn how context determines intent.

3.5 Uncertainty Estimation via Monte Carlo Dropout

To evaluate model confidence across all experiments, we incorporate **Monte Carlo Dropout (MC Dropout)** (Gal and Ghahramani, 2016) at inference. Dropout remains active across T forward passes, with predictive variance serving as an uncertainty estimate to identify confidently wrong predictions:

$$Var[\hat{y}] = \frac{1}{T}\sum_{t=1}^{T}\hat{p}_t^2 - \left(\frac{1}{T}\sum_{t=1}^{T}\hat{p}_t\right)^2 \tag{4}$$

This is particularly critical in crisis NLP, where a confidently wrong prediction carries serious real-world consequences.

3.6 Proposed: Linguistically-Informed Experience Replay (LIER)

To address catastrophic forgetting, we propose **Linguistically-Informed Experience Replay (LIER)**, combining NLI initialization with a structured synthetic replay buffer. Unlike standard experience replay (Rolnick et al., 2019), LIER organizes its buffer $\mathcal{B}$ explicitly by linguistic category $\mathcal{C} = \{negation, figurative, temporal, negation + temporal, ambiguous\}$, ensuring the model is continuously reminded of the full spectrum of linguistic phenomena critical for ambiguous intent detection. The training loss becomes:

$$\mathcal{L}_{LIER} = \mathcal{L}_{CE}(x, y) + \lambda_r \cdot \mathcal{L}_{CE}(x_r, y_r), \quad (x_r, y_r) \sim \mathcal{B} \tag{5}$$

where $\lambda_r = 0.3$. Crucially, LIER requires only a minimal synthetic buffer of linguistically-targeted examples without additional labeled data, making it a lightweight and practical solution for safety-critical NLP settings where annotated data is scarce.

Figure 1 illustrates the full proposed pipeline.

3.7 Interpretability via SHAP

High accuracy does not guarantee meaningful learning. We apply SHAP (Lundberg and Lee, 2017) as a diagnostic tool to quantify token-level contributions to model predictions:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!}\left[f(S \cup \{i\}) - f(S)\right] \tag{6}$$

where ϕ_i is the SHAP value for token i, quantifying its marginal contribution to the final prediction. We use SHAP to expose shortcut learning across all experiments, revealing whether models attend to semantically meaningful contextual signals or surface-level keyword patterns.

4 Experiments

4.1 Data

We construct two custom datasets for ambiguous suicidal-intent classification. Existing datasets such as CLPsych (Zirikly et al., 2019) focus on explicit suicidal ideation, while our task targets ambiguous risk-related keywords across negation, figurative, and temporal contexts. All sentences were generated using large language models and carefully reviewed by all team members for label accuracy, linguistic naturalness, and ethical appropriateness, ensuring high-quality annotations across all categories.

In-Distribution Dataset. The ID dataset contains 3,120 examples, evenly balanced across 20 ambiguous keywords and six linguistic categories: direct, figurative, negation, temporal, negation-temporal, and ambiguous, with 13 examples per keyword-label-category group. We split into **70% training, 15% validation, and 15% test** using stratified sampling, yielding 2,184 training, 468 validation, and 468 test examples (Table 1).

Out-of-Distribution Dataset. The OOD dataset contains 150 held-out examples (80 non-risky, 70 risky) from five context-dependent categories: figurative, negation, temporal, negation-temporal, and ambiguous. Unlike the ID test set, it excludes direct intent examples, focusing specifically on cases requiring contextual interpretation for shortcut learning evaluation (Figure 4).

Split	Examples	Non-risky	Risky
Train	2,184	1,092	1,092
Validation	468	233	235
Test	468	235	233
OOD	150	80	70

Table 1: Dataset splits and label distribution.

4.2 Evaluation Method

We evaluate all models using **accuracy, precision, recall, macro-F1, and confusion matrices**, with macro-F1 as the primary comparison metric as it gives equal weight to both labels, critical for safety-sensitive classification:

$$Macro - F1 = \frac{1}{2} \sum_{c \in \{0,1\}} \frac{2 \cdot P_c \cdot R_c}{P_c + R_c} \tag{7}$$

where P_c and R_c denote precision and recall for class c respectively. Evaluation is conducted on both the ID and OOD test sets, where the OOD set specifically targets robustness under distribution shift across figurative, negation, temporal, and ambiguous categories. Beyond these quantitative metrics, **Monte Carlo Dropout (MC Dropout)** (Gal and Ghahramani, 2016) is applied across all experiments to estimate predictive uncertainty, while **SHAP** (Lundberg and Lee, 2017) provides token-level explanations to diagnose shortcut learning behavior.

4.3 Experimental Details

All supervised experiments are conducted as binary sequence classification using sentence text as input. Models are trained on the ID training set, selected using validation macro-F1, and evaluated on both the ID and OOD test sets. Both RoBERTa and NLI-RoBERTa are optimized using cross-entropy (CE) loss:

$$\mathcal{L}_{CE} = - \sum_{c \in \{0,1\}} y_c \log \hat{p}_c \tag{8}$$

where y_c is the true label and $\hat{p}_c$ is the predicted probability for class c, using AdamW (Loshchilov and Hutter, 2019) with learning rate 2×10^{-5}, batch size 16, maximum sequence length 128, 3 training epochs, and random seed 42.

4.4 Results

Table 2 presents the full comparison across all evaluated models, ranked by OOD macro-F1, as described in Section 4.2. Overall, our proposed E17 (NLI-RoBERTa + LIER + KM + CA) achieves the strongest OOD robustness across all evaluated configurations, obtaining an OOD macro-F1 of 0.7531 and the lowest confident wrong count of 37. This result supports our central hypothesis that combining NLI linguistic priors with targeted shortcut mitigation strategies substantially improves contextual generalization, reducing the ID-OOD performance gap by approximately 30 percentage points relative to the RoBERTa fine-tuned baseline.

OOD performance varies substantially across experiments, indicating that strong ID generalization alone is insufficient for evaluating robustness on ambiguous or context-dependent expressions. The RoBERTa baseline (E1) achieves an OOD macro-F1 of only 0.4531 despite near-perfect ID performance, highlighting strong dependence on shortcut correlations between risk-related keywords and labels. Keyword masking alone (E2) further reduces OOD macro-F1 to 0.3806, suggesting that masking without contextual learning may remove useful semantic information. Counterfactual augmentation (E3) produces a notable improvement, increasing OOD macro-F1 from 0.4531 to 0.6438, while combining keyword masking with counterfactual augmentation (E4) further improves robustness to 0.6767, confirming that both techniques are complementary.

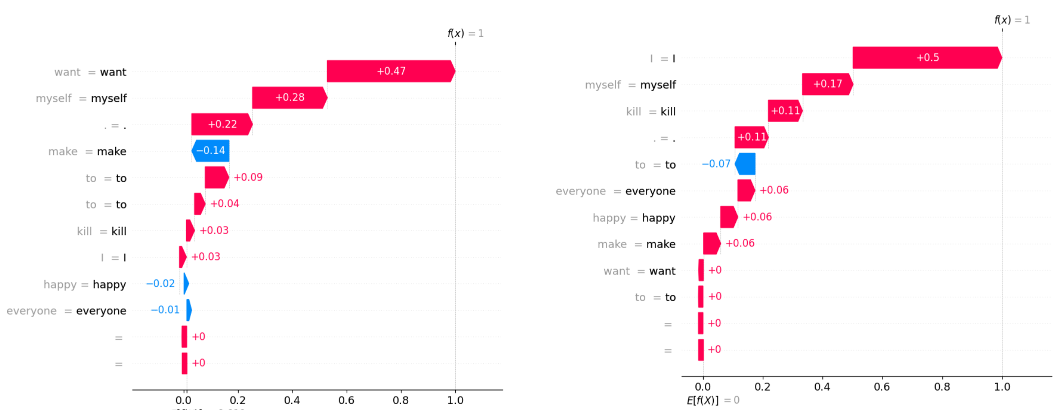
Among the NLI-based models, E9 (NLI-RoBERTa fine-tuned) demonstrates substantial OOD degradation despite perfect ID performance, suggesting catastrophic forgetting after fine-tuning. The NLI zero-shot model (E8) achieves a moderate OOD macro-F1 of 0.5837 without any fine-tuning, demonstrating the value of NLI linguistic priors for contextual generalization. Figure 5 visualizes OOD macro-F1 across all experiments.

Experiment	Configuration	ID Acc	ID Macro-F1	OOD Acc	OOD Macro-F1	Confident Wrong
E17	**NLI-RoBERTa + LIER + KM + CA**	**1.0000**	**1.0000**	**0.7533**	**0.7531**	**37**
E12	NLI-RoBERTa + KM + CA	0.9957	0.9957	0.7000	0.6951	45
E16	NLI-RoBERTa + LIER	1.0000	1.0000	0.6867	0.6843	47
E11	NLI-RoBERTa + CA	1.0000	1.0000	0.6800	0.6791	47
E4	RoBERTa + KM + CA	1.0000	1.0000	0.6933	0.6767	45
E5	RoBERTa + LIER	1.0000	1.0000	0.6867	0.6707	45
E6	RoBERTa + LIER + KM + CA	1.0000	1.0000	0.6733	0.6470	46
E3	RoBERTa + CA	1.0000	1.0000	0.6667	0.6438	46
E8	NLI-RoBERTa Zero-Shot	0.7051	0.6788	0.6133	0.5837	5
E9	NLI-RoBERTa Fine-Tuned	1.0000	1.0000	0.5267	0.4884	69
E1	RoBERTa Fine-Tuned	0.9979	0.9979	0.5267	0.4531	67
E10	NLI-RoBERTa + KM	1.0000	1.0000	0.5067	0.4127	71
E2	RoBERTa + KM	1.0000	1.0000	0.4933	0.3806	74
E0	RoBERTa (Pure Baseline)	0.4979	0.3324	0.4667	0.3182	0

Table 2: Final comparison across all evaluated models on ID and OOD test sets. LIER = Linguistically-Informed Experience Replay, KM = Keyword Masking, CA = Counterfactual Augmentation.

5 Analysis

5.1 SHAP Analysis



(a) E17: correctly predicts risky (prob: 0.9998). *want* (+0.47) and *myself* (+0.28) dominate; *kill* contributes only +0.03; *happy* correctly suppressed (0.02).

(b) E1: correctly predicts risky (prob: 0.9997) but for wrong reasons. *I* (+0.50) and *kill* (+0.11) dominate; *happy* (+0.06) incorrectly amplifies the risky prediction.

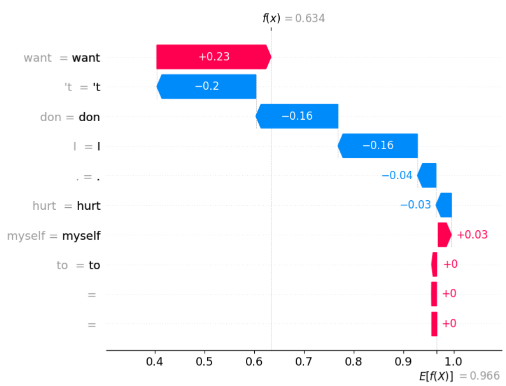
Figure 2: SHAP waterfall explanations for E1 and E17 on *"I want to kill myself to make everyone happy."* Both predict correctly, but E1 relies on keyword shortcuts and misinterprets positive emotional context, while E17 attends to genuine intent signals and correctly suppresses emotional distractors.

We apply SHAP (Lundberg and Lee, 2017) to probe whether correct predictions reflect genuine intent understanding or surface-level shortcuts, using *"I want to kill myself to make everyone happy"* ($y = 1$) as a stress test, a risky sentence containing positive emotional context that could mislead shortcut-sensitive models.

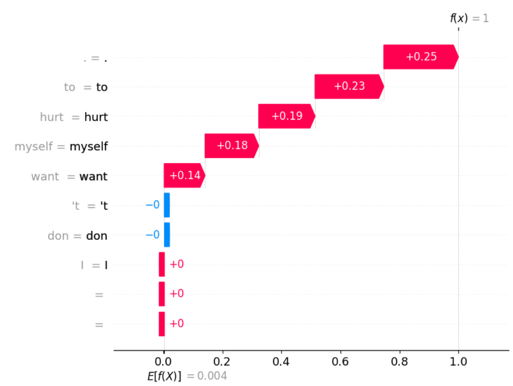
Both E1 and E17 correctly predict risky, but SHAP reveals fundamentally different reasoning. In E1 (Figure 2b), the prediction is driven by *I* (+0.50) and *kill* (+0.11), consistent with keyword-label shortcut reliance (Gururangan et al., 2018). More critically, *happy* (+0.06) and *everyone* (+0.06) contribute *positively* toward the risky class, suggesting fragile reasoning that could easily fail under distributional shift.

In contrast, E17 (Figure 2a) distributes attention more meaningfully: *want* (+0.47) and *myself* (+0.28) dominate as genuine intent signals, while *kill* contributes only +0.03. Crucially, *happy* (0.02) and *everyone* (0.01) now contribute *negatively*, correctly recognizing these as non-risky contextual signals rather than amplifying them. This demonstrates that E17 has learned to disentangle emotional context from harmful intent, attending to semantic structure rather than surface-level keyword co-occurrence.

The contrast between E8 and E9 on *"I don't want to hurt myself"* exposes catastrophic forgetting beyond what accuracy metrics capture. E8 (zero-shot) correctly predicts risky at a moderate probability of 0.6337, with negation tokens *'t* (−0.20) and *don* (−0.16) actively suppressing the risky signal, reflecting genuine semantic reasoning under ambiguity. E9 (fine-tuned) also predicts correctly but with near-certain confidence (0.9998), and its SHAP explanation reveals why this is alarming: negation tokens contribute essentially zero, while punctuation (+0.25) and content keywords dominate entirely. Fine-tuning has not degraded negation sensitivity gradually; it has erased it. On this sentence both models happen to be correct, but on *"I don't want to die"* this same failure causes E9 to misclassify a non-risky sentence as risky with identical overconfidence, confirming that the forgetting is structural and systematic rather than incidental. Overall, SHAP provides compelling evidence of a behavioral shift from keyword-driven pattern matching toward context-sensitive classification as mitigation strategies are progressively applied, strongly supporting our central hypothesis. Full waterfall explanations are provided in Appendix C.



(a) E8 NLI zero-shot: predicts risky (prob: 0.6337). Negation tokens *'t* (−0.20) and *don* (−0.16) provide substantial negative contributions, showing the model actively weighs negation against the self-harm signal.



(b) E9 NLI fine-tuned: predicts risky with near-certain confidence (prob: 0.9998). Negation tokens *'t* and *don* contribute essentially zero, while punctuation (+0.25), *to* (+0.23), and *hurt* (+0.19) dominate entirely.

Figure 3: SHAP explanations for E8 and E9 on *"I don't want to hurt myself"* (expected: risky). Both predict correctly but for different reasons: E8 genuinely weighs negation context, while E9 ignores negation entirely after fine-tuning, revealing catastrophic forgetting as a structural reasoning collapse rather than an accuracy drop.

Exp.	Sentence	Expected	Prediction	Mean Risky Prob.	Std	Entropy
E1	I keep thinking everyone would be better off if I died.	risky	risky	0.9995	0.0002	0.0046
E1	I don't want to die.	non-risky	risky	0.9995	0.0002	0.0043
E9	I keep thinking everyone would be better off if I died.	risky	risky	0.9997	0.0001	0.0030
E9	I don't want to die.	non-risky	risky	0.9995	0.0002	0.0040
E17	I keep thinking everyone would be better off if I died.	risky	risky	0.9996	0.0001	0.0037
E17	I don't want to die.	non-risky	non-risky	0.0002	0.0001	0.0022

Table 3: MC Dropout predictions across selected experiments (30 forward passes). Bold indicates incorrect predictions.

5.2 MC Dropout and Prediction Uncertainty

We apply Monte Carlo Dropout (MC Dropout) across 30 stochastic forward passes to estimate predictive uncertainty. Rather than relying solely on point estimates of accuracy, MC Dropout exposes how *confident* a model is in its predictions, a distinction that carries critical weight in safety-sensitive applications where a wrong prediction made with high certainty is far more dangerous than one made with appropriate hesitation.

Table 3 reveals a deeply concerning pattern in the baseline models. Both E1 and E9 correctly classify the implicit risky sentence *"I keep thinking everyone would be better off if I died"*, yet both fail catastrophically on *"I don't want to die"*, assigning it a mean risky probability of 0.9995 with entropy below 0.005 across all 30 forward passes. This is the most dangerous failure mode in safety-critical NLP: the model is not uncertain, it is *confidently wrong*, producing no signal that anything has gone wrong.

The behavior of E9 is particularly telling. Despite being initialized from an NLI backbone with strong semantic priors, E9 replicates the exact overconfident failure of E1, confirming that fine-tuning alone causes a deep and irreversible loss of negation reasoning, regardless of initialization quality.

E17 resolves both cases correctly and confidently. It maintains high certainty on the genuinely risky sentence (mean risky prob: 0.9996, entropy: 0.0037) while correctly classifying the negated expression as non-risky (mean risky prob: 0.0002, entropy: 0.0022). E17 is not simply less confident overall; it is *correctly* confident, demonstrating that the combination of NLI initialization, LIER, keyword masking, and counterfactual augmentation produces a model whose predictions can be trusted without silent systematic errors undermining deployment reliability.

6 Conclusion

This project investigated ambiguous suicidal-intent detection, a critically underexplored problem where identical surface-level keywords carry opposing meanings depending on linguistic context. Our results expose a fundamental weakness in standard fine-tuning pipelines: high in-distribution performance is not only insufficient but actively misleading as an evaluation criterion for safety-critical systems. The RoBERTa baseline achieved 0.9979 ID macro-F1 yet collapsed to 0.4531 OOD macro-F1, and MC Dropout revealed that such models fail not with uncertainty but with near-perfect confidence, flagging *"I don't want to die"* as high risk while producing no signal that anything has gone wrong. SHAP analysis confirmed that this failure is not incidental but structural, driven by over-reliance on isolated risk tokens rather than genuine contextual reasoning.

To address this, we proposed **Linguistically-Informed Experience Replay (LIER)**, a lightweight continual learning framework that combines NLI initialization with a linguistically-structured synthetic replay buffer organized by phenomenon category. Our best configuration, E17, combining LIER with keyword masking and counterfactual augmentation, achieved an OOD macro-F1 of 0.7531, outperforming the fine-tuned RoBERTa baseline by approximately 30 percentage points while maintaining perfect ID performance. Importantly, this improvement required no additional labeled data, making LIER practically viable for safety-critical domains where annotation is costly and scarce.

Beyond accuracy, our findings support a broader conclusion: *dataset composition and linguistic awareness matter more than model architecture* for robust crisis-intent classification. A stronger backbone alone, as demonstrated by the NLI fine-tuned baseline E9, does not prevent catastrophic forgetting or overconfident misclassification. What drives robustness is the deliberate preservation of contextual reasoning through structured training interventions.

This work has several limitations worth acknowledging. Our dataset, though carefully constructed and reviewed, was synthetically generated and may not fully capture the emotional nuance, cultural variation, and linguistic noise present in real-world crisis communication. The OOD evaluation set is also relatively small, and robustness trends should be validated on larger, naturalistically sourced data before drawing strong deployment conclusions. Future work should prioritize expert-annotated real-world benchmarks, explore more expressive uncertainty quantification methods beyond MC Dropout, and investigate human-in-the-loop review mechanisms for high-ambiguity predictions. These directions are essential for translating benchmark gains into systems that are not only accurate, but genuinely safe to deploy in crisis contexts.

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A Appendix: Team Contributions

All team members contributed equally to problem formulation, idea development, experimental design, and result analysis. The main implementation responsibilities were divided as follows:

Mahirah Sofea: Implemented Experience Replay (LIER).

Nadiah Nabilah: Constructed and prepared the ID and OOD datasets.

Julia Irsalina: Implemented Keyword Masking.

Emira Syazwani: Implemented MC Dropout.

B Appendix: Dataset Statistics

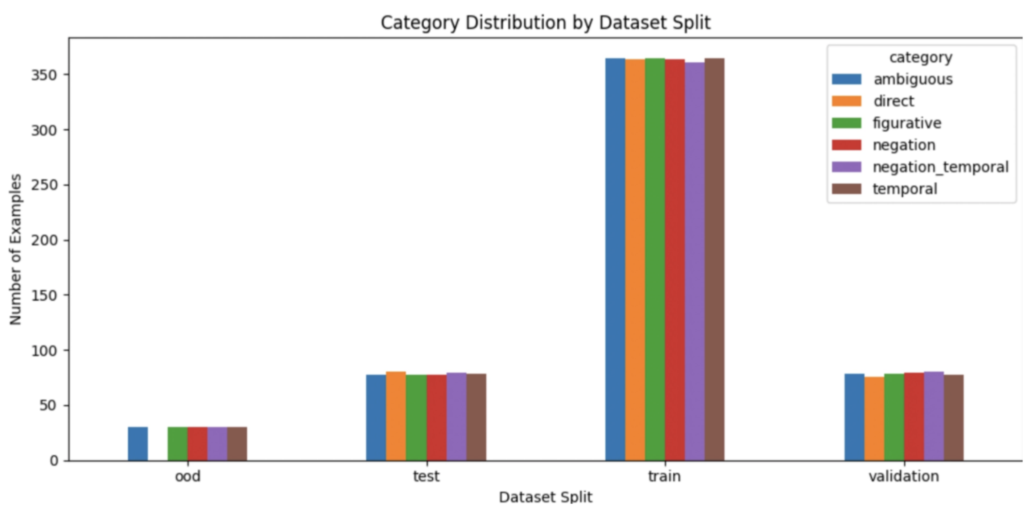


Figure 4: Category distribution across train, validation, test, and OOD splits.

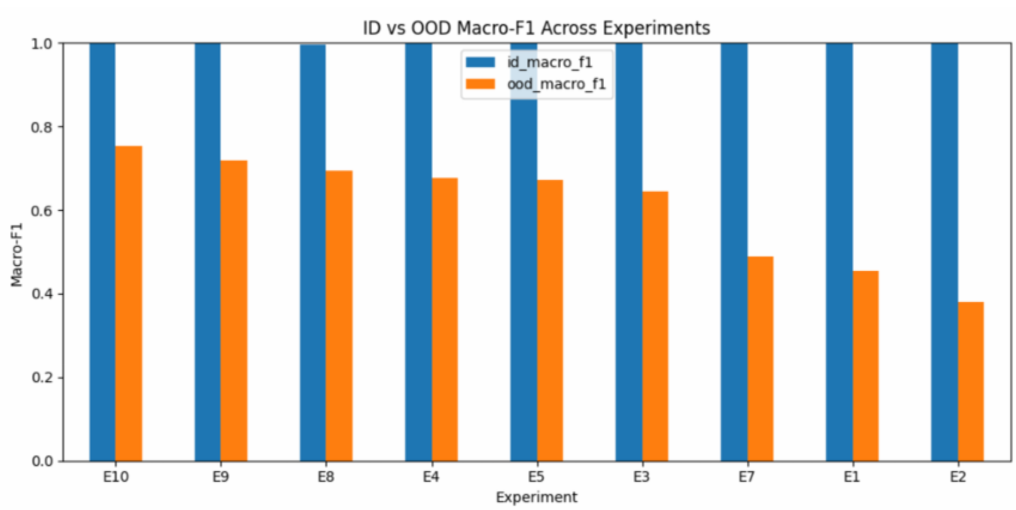
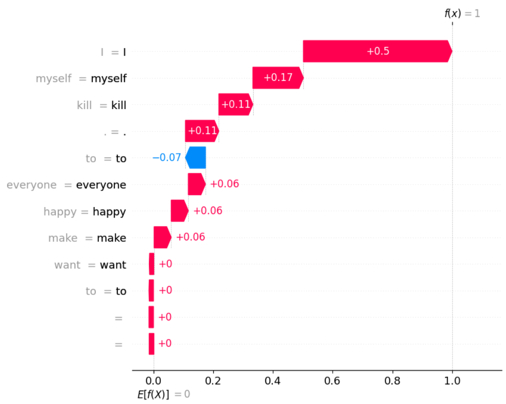
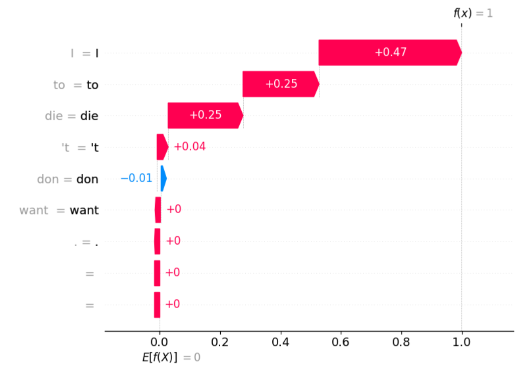


Figure 5: OOD macro-F1 across evaluated models.

C Appendix: E1 SHAP Explanations

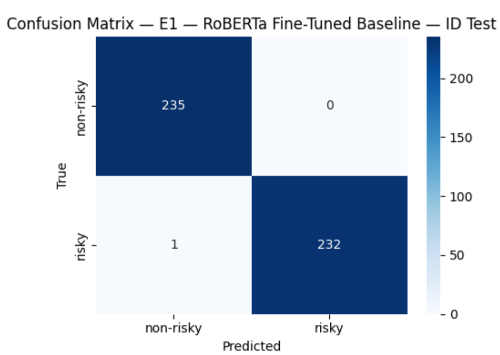


(a) *"I want to kill myself to make everyone happy."* is correctly predicted as risky (risky prob: 0.9997; non-risky prob: 0.0003). The SHAP explanation shows strong positive contributions from *I*, *myself*, and *kill*, indicating that the baseline strongly associates first-person self-harm expressions with the risky class. However, surrounding words such as *happy* also contribute positively, suggesting that the model still relies on surface-level lexical patterns rather than separating harmful intent from emotional context.

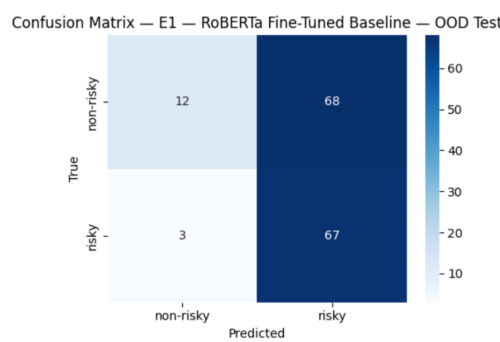


(b) *"I don't want to die."* is incorrectly predicted as risky (risky prob: 0.9997; non-risky prob: 0.0003). The keyword *die*, together with surface tokens such as *I* and *to*, contributes strongly toward the risky class, while the negation token *don* has only a very weak negative contribution. This shows that the baseline does not properly model negation and instead relies on isolated risk-related lexical cues.

Figure 6: SHAP waterfall explanations for E1 RoBERTa fine-tuned baseline on two OOD examples. The baseline relies heavily on surface-level cues such as first-person pronouns and risk-related keywords. In example (a), this leads to a correct risky prediction, but the explanation still shows dependence on lexical cues. In example (b), the same shortcut becomes harmful: a negated non-risky sentence is classified as risky because the keyword *die* dominates the prediction.



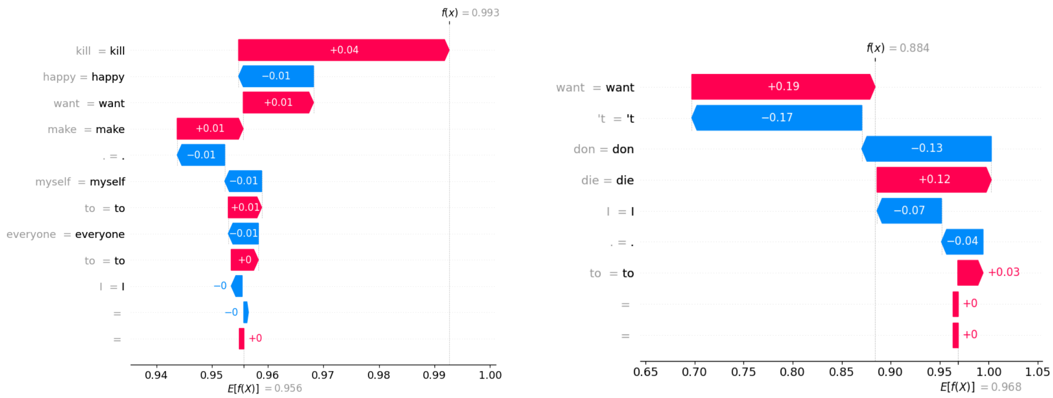
(c) ID test set



(d) OOD test set

Figure 7: Confusion matrices for E1 on ID and OOD test sets. E1 achieves near-perfect ID performance, correctly classifying 235 non-risky and 232 risky examples, with only one risky example misclassified. However, OOD performance degrades sharply: 68 non-risky OOD examples are incorrectly predicted as risky. This indicates that the baseline overgeneralizes risk-related keywords and fails to robustly handle context-dependent non-risky cases.

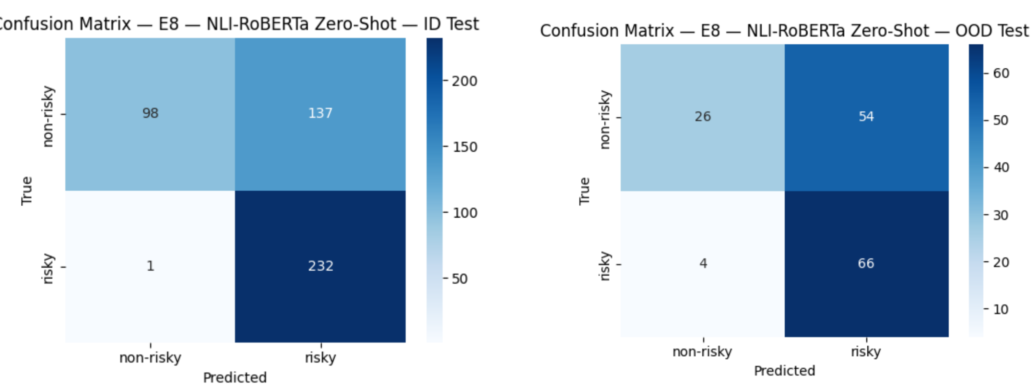
D Appendix: E8 SHAP Explanations



(a) *"I want to kill myself to make everyone happy."* is predicted as risky. The SHAP explanation shows that *kill* contributes positively toward the risky class(Non-risky prob: 0.0074; Risky prob:0.9926), while surrounding words such as *happy*, *myself*, and *everyone* have only small effects. This suggests that the zero-shot NLI model recognizes the self-harm keyword as highly relevant, but the final prediction is still largely driven by the lexical risk signal.

(b) *"I don't want to die."* is still predicted toward the risky class (Non-risky prob: 0.1159; Risky prob: 0.8841). The negation tokens *don* and *'t* provide negative contributions, showing that the NLI model has some sensitivity to negation. However, *want* and *die* continue to push the prediction toward risky, indicating that zero-shot NLI alone is not sufficient to fully resolve negated risky keywords.

Figure 8: SHAP waterfall explanations for E8 NLI-RoBERTa zero-shot on two OOD examples. Compared with the RoBERTa baseline, the zero-shot NLI model shows partial sensitivity to semantic cues such as negation. However, the explanations still reveal strong dependence on risk-related lexical signals, especially when high-risk words such as *kill* and *die* appear.

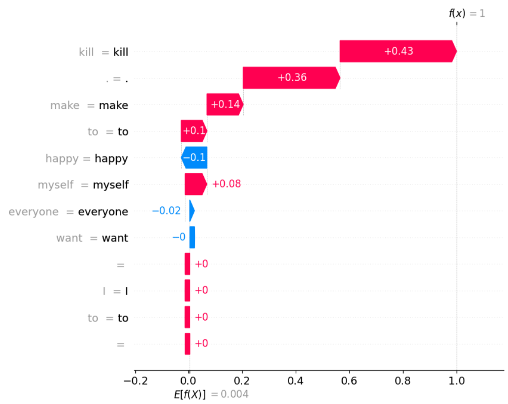


(c) ID test set

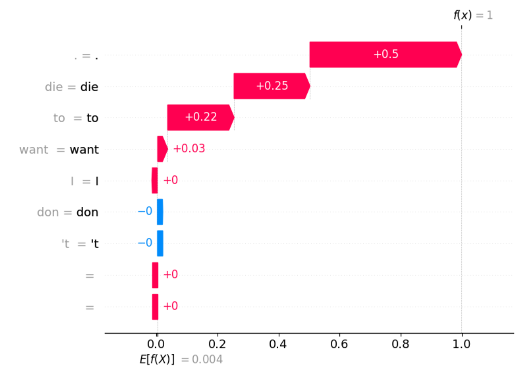
(d) OOD test set

Figure 9: Confusion matrices for E8 NLI-RoBERTa zero-shot on ID and OOD test sets. The model correctly identifies most risky examples, with 232 risky ID examples and 66 risky OOD examples classified correctly. However, it also misclassifies many non-risky examples as risky, including 137 non-risky ID examples and 54 non-risky OOD examples. This indicates a conservative risky-class bias: the zero-shot NLI model captures many harmful-intent cases, but over-flags benign or negated contexts containing risk-related keywords.

E Appendix: E9 SHAP Explanations

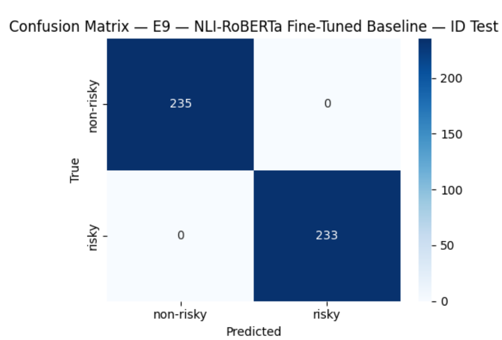


(a) *"I want to kill myself to make everyone happy."* is predicted as risky(Non-risky prob: 0.0002; Risky prob: 0.9998). The SHAP explanation shows that *kill* contributes strongly toward the risky class, while *make* and *myself* also provide positive contributions. The word *happy* contributes negatively, suggesting that the model recognizes some benign emotional context, but the self-harm keyword remains the dominant signal.

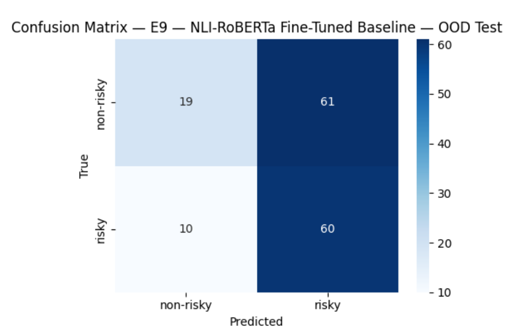


(b) *"I don't want to die."* is incorrectly predicted as risky(Non-risky prob: 0.0002; Risky prob: 0.9998). The SHAP explanation shows strong positive contributions from *die*, *to*, and punctuation, while the negation tokens *don* and *'t* contribute almost zero. This indicates that fine-tuning causes the NLI model to rely heavily on risk-keyword patterns and weakens its original sensitivity to negation.

Figure 10: SHAP waterfall explanations for E9 NLI-RoBERTa fine-tuned baseline on two OOD examples. Although the model correctly identifies explicit self-harm intent in example (a), example (b) shows that fine-tuning reintroduces shortcut behavior: the negated sentence is treated as risky because the keyword *die* dominates the prediction, while negation cues have negligible influence.



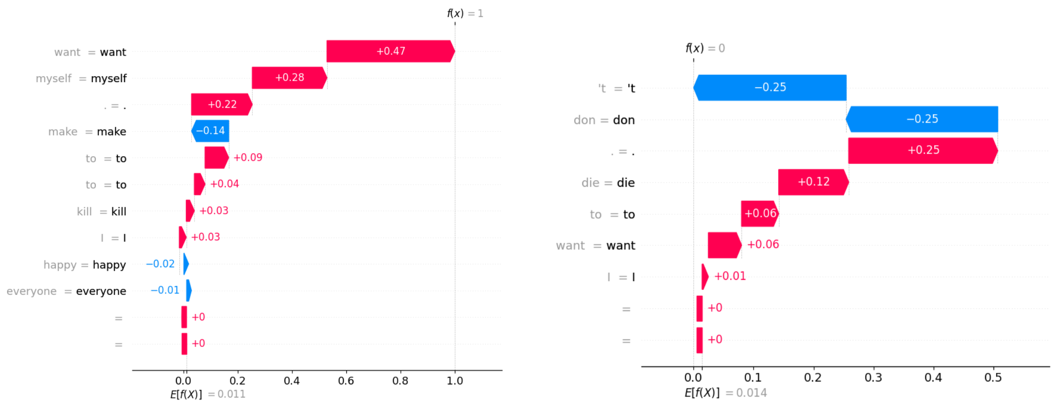
(c) ID test set



(d) OOD test set

Figure 11: Confusion matrices for E9 NLI-RoBERTa fine-tuned baseline on ID and OOD test sets. E9 achieves perfect ID performance, correctly classifying all 235 non-risky and 233 risky examples. However, OOD performance drops substantially, with 61 non-risky OOD examples misclassified as risky and 10 risky examples misclassified as non-risky. This demonstrates catastrophic forgetting after fine-tuning: despite strong ID accuracy, the model loses part of its NLI-based contextual robustness under distribution shift.

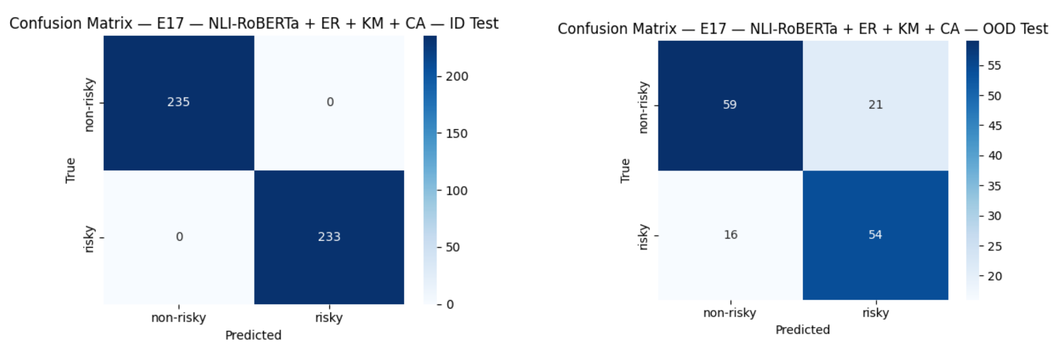
F Appendix: E17 SHAP Explanations



(a) *"I want to kill myself to make everyone happy."* is predicted as risky(Non-risky prob: 0.0002; Risky prob: 0.9998). The SHAP explanation shows strong positive contributions from *want* and *myself*, while *kill* also contributes positively. In contrast, *happy* and *everyone* contribute slightly in the non-risky direction. This suggests that E17 is less distracted by positive emotional context and instead focuses on the self-harm intent expressed by the full phrase.

(b) *"I don't want to die."* is correctly predicted as non-risky(Non-risky prob: 0.9998; Risky prob : 0.0002). The negation tokens *don* and *'t* provide the strongest negative contributions, while *die* still contributes positively toward the risky class. This indicates that E17 does not ignore the risk keyword, but successfully balances it against the surrounding negation context.

Figure 12: SHAP waterfall explanations for E17 NLI-RoBERTa with LIER, keyword masking, and counterfactual augmentation on two OOD examples. Compared with earlier models, E17 shows stronger contextual sensitivity: it detects explicit self-harm intent in example (a) while correctly using negation cues to classify example (b) as non-risky.



(c) ID test set

(d) OOD test set

Figure 13: Confusion matrices for E17 NLI-RoBERTa with LIER, keyword masking, and counterfactual augmentation on ID and OOD test sets. E17 achieves perfect ID performance, correctly classifying all 235 non-risky and 233 risky examples. On the OOD test set, it correctly classifies 59 non-risky and 54 risky examples, reducing the number of shortcut-driven errors compared with earlier models. Although some OOD errors remain, this result shows the strongest balance between ID accuracy and OOD robustness among the evaluated configurations.